

Experiment - 4

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Branch: MCA

Section/Group: 24MCA-1B

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Subject Name: Backend Technologies Lab

Subject Code: 24CAP-709

Aim of Practical:

To design and implement a **University Course Management Application** that allows CRUD operations for departments, courses, and modules.

Objectives:

1. Understand MVC structure in Node.js.
2. Learn to implement RESTful APIs.
3. Practice using MongoDB with Mongoose.
4. Build frontend integration with backend APIs.
5. Gain experience in modular coding and reusability.

Project Structure:

University-course-app/

├─ controllers/

| └─ departmentController.js

| └─ courseController.js

| └─ moduleController.js

├─ models/

| └─ department.js

| └─ course.js

| └─ module.js

├─ routes/

| └─ departmentRoutes.js

- | └─ courseRoutes.js
- | └─ moduleRoutes.js
- | └─ config/
- | └─ database.js
- | └─ frontend/
- | └─ index.html
- | └─ App.js
- | └─ package.json

Input/Code: (index.html)

```
<!doctype html>

<html lang="en">

<head>

  <meta charset="utf-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1" />

  <title>University Dashboard</title>

  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;600;700&display=swap" rel="stylesheet">

  <style>

    :root{

      --bg:#f7fafc; --card:#ffffff; --muted:#6b7280;

      --accent-a:#0ea5a4; --accent-b:#3b82f6; --accent-c:#f97316;

      --glass: rgba(255,255,255,0.6);

      font-family: 'Inter', system-ui, -apple-system, 'Segoe UI', Roboto, 'Helvetica Neue', Arial;

    }

    *{box-sizing:border-box}

    body{margin:0;background:linear-gradient(180deg,#eef2ff 0%,var(--bg) 100%);color:#0f172a}
```



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```
header{display:flex;align-items:center;justify-content:space-between;padding:18px 28px;background:transparent}

.brand{display:flex;gap:14px;align-items:center}

.logo{width:48px;height:48px;border-radius:10px;background:linear-gradient(135deg,var(--accent-a),var(--accent-b));display:flex;align-items:center;justify-content:center;color:white;font-weight:700;box-shadow:0 6px 18px rgba(59,130,246,0.12)}

h1{font-size:20px;margin:0}

.subtitle{font-size:13px;color:var(--muted)}

/* Layout */

.container{

display: grid;

grid-template-columns: repeat(3, 1fr);

gap: 24px;

justify-content: center; /* centers the grid horizontally */

align-items: start; /* keeps top alignment */

max-width: 1200px;

margin: 40px auto; /* adds even margin around */

padding: 0 20px;

}

.panel{background:var(--card);border-radius:12px;padding:16px;box-shadow:0 8px 24px rgba(15,23,42,0.06);min-height:520px;display:flex;flex-direction:column}

/* Panel header */

.panel-head{display:flex;align-items:center;justify-content:space-between;margin-bottom:12px}

.panel-title{display:flex;gap:10px;align-items:center}

.dot{width:12px;height:12px;border-radius:8px}

.dot.a{background:var(--accent-a)} .dot.b{background:var(--accent-b)} .dot.c{background:var(--accent-c)}

.panel-actions{display:flex;gap:8px}
```



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```
.btn{padding:8px 12px;border-radius:8px;border:0;font-weight:600;cursor:pointer}

.btn.ghost{background:transparent;border:1px solid #eef2ff;color:var(--muted)}

.btn.primary{background:linear-gradient(90deg,var(--accent-b),var(--accent-a));color:white}

.btn.warn{background:var(--accent-c);color:white}

/* Form */

form.grid{display:grid;grid-template-columns:1fr 1fr;gap:10px;margin-bottom:12px}

input,select{padding:10px;border-radius:8px;border:1px solid #e6edf3;font-size:14px}

.full{grid-column:span 2}

/* Table area */

.table-wrap{overflow:auto;flex:1}

table{width:100%;border-collapse:collapse;font-size:14px}

thead th{position:sticky;top:0;background:linear-gradient(90deg,rgba(255,255,255,0.9),rgba(255,255,255,0.8));padding:10px;text-align:left;border-bottom:1px solid #eef2ff}

tbody td{padding:10px;border-bottom:1px dashed #f1f5f9}

.actions{display:flex;gap:8px}

.pill{display:inline-block;padding:6px 8px;border-radius:999px;font-size:12px;background:#f1f5f9;color:var(--muted)}

/* Responsive */

@media (max-width:1100px){.container{grid-template-columns:1fr;max-width:760px}}

/* Small helpers */

.search{display:flex;gap:8px;align-items:center}

.compact{font-size:13px;color:var(--muted)}

.footer-note{font-size:12px;color:var(--muted);margin-top:10px}

/* subtle card separator for mobile stacking */
```



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@media (max-width:1100px){.panel{min-height:360px}}

</style>

</head>

<body>

<header>

<div class="brand">

<div class="logo">U</div>

<div>

<h1>University Dashboard</h1>

<div class="subtitle"> overview — Departments · Courses · Modules</div>

</div>

</div>

<div class="compact">Connected to: /api/ (mock)</div>

</header>

<main class="container">

<!-- Departments -->

<section class="panel" id="panel-departments">

<div class="panel-head">

<form id="form-dep" class="grid" onsubmit="createDepartment(event)">

<input id="dep_id" placeholder="Dept ID (eg. D001)" required />

<input id="dep_name" placeholder="Department Name" required />

<input id="location" placeholder="Location" />

<div style="display:flex;gap:8px;align-items:center">

<input id="search_dep_id" placeholder="Search by ID" />

<button type="button" class="btn ghost" onclick="searchDepartment()">Search</button>

</div>

```
<div class="full" style="display: flex; gap: 8px; justify-content: flex-end">
```

```
  <input id="update_dep_id" placeholder="ID to update" style="flex: 1" />
```

```
  <input id="update_dep_name" placeholder="New name" style="flex: 1" />
```

```
  <button type="button" class="btn warn" onclick="updateDepartment()">Update</button>
```

```
</div>
```

```
<div class="full" style="display: flex; justify-content: flex-end">
```

```
  <button type="submit" class="btn primary">Add Department</button>
```

```
</div>
```

```
</form>
```

```
<div class="footer-note">Tip: Click Delete to remove a department. Use Update section to modify names.</div>
```

```
</section>
```

```
<!-- Courses -->
```

```
<section class="panel" id="panel-courses">
```

```
  <div class="panel-head">
```

```
    <div class="panel-title"><div class="dot b"></div><div><strong>Courses</strong></div><div class="compact">Associate  
    courses to departments</div></div></div>
```

```
  <div class="panel-actions">
```

```
    <button class="btn ghost" id="refresh-course">Refresh</button>
```

```
    <button class="btn primary" id="export-course-xlsx">Export XLSX</button>
```

```
  </div>
```

```
<div class="full" style="display: flex; gap: 8px; justify-content: flex-end">
```

```
  <input id="update_course_id" placeholder="ID to update" style="flex: 1" />
```

```
  <input id="update_course_name" placeholder="New name" style="flex: 1" />
```

```
  <button type="button" class="btn warn" onclick="updateCourse()">Update</button>
```

```
</div>
```

```
<div class="full" style="display: flex; justify-content: flex-end">
```

```
  <button type="submit" class="btn primary">Add Course</button>
```



</form>

<div class="table-wrap">

<table id="courseTable">

<thead>

<tr><th>ID</th><th>Dept ID</th><th>Name</th><th>Actions</th></tr>

</thead>

<tbody></tbody>

</table>

</div>

<div class="footer-note">Tip: Keep Dept ID consistent with Departments. Use Refresh after backend changes.</div>

</section>

<!-- Modules -->

<section class="panel" id="panel-modules">

<div class="panel-head">

<div class="panel-title"><div class="dot c"></div><div>Modules<div class="compact">Course modules / subjects</div></div></div>

<div class="panel-actions">

<button class="btn ghost" id="refresh-mod">Refresh</button>

<button class="btn primary" id="export-mod-xlsx">Export XLSX</button>

</div>

</div>

<form id="form-mod" class="grid" onsubmit="createModule(event)">

<input id="module_c_id" placeholder="Course ID (eg. C101)" required />

<input id="mod_id" placeholder="Module ID (eg. M01)" required />

<input id="mod_name" placeholder="Module Name" required />

<div style="display:flex;gap:8px;align-items:center">

<input id="search_mod_id" placeholder="Search Module ID" />

Search</button>

</div>

<div class="full" style="display:flex;gap:8px;justify-content:flex-end">

<input id="update_mod_id" placeholder="ID to update" style="flex:1" />

<input id="update_mod_name" placeholder="New name" style="flex:1" />

<button type="button" class="btn warn" onclick="updateModule()">Update</button>

</div>

<div class="full" style="display:flex;justify-content:flex-end">

<button type="submit" class="btn primary">Add Module</button>

</div>

</form>

<div class="table-wrap">

<table id="moduleTable">

<thead>

<tr><th>ID</th><th>Course ID</th><th>Name</th><th>Actions</th></tr>

</thead>

<tbody></tbody>

</table>

</div>

<div class="footer-note">Tip: Modules link to Course ID. Use Update to rename modules.</div>

</section>

</main>

<!-- Minimal JS (same API endpoints as your original file) -->

<script>

```
    async function createDepartment(e){e.preventDefault();const dep_id=document.getElementById('dep_id').value.trim();const
    dep_name=document.getElementById('dep_name').value.trim();const
    location=document.getElementById('location').value.trim();if(!dep_id||!dep_name){flash('Provide ID and
    Name');return}try{await fetch('/api/departments',{method:'POST',headers:{'Content-
    Type':'application/json'},body:JSON.stringify({dep_id,dep_name,location})});fetchDepartments();flash('Added')}catch(err){cons
    ole.error(err);flash('Error')} }
```



```
async function fetchDepartments() {try {const res=await fetch('/api/departments');const data=await res.json();const tbody=document.querySelector('#departmentTable tbody');tbody.innerHTML="";data.forEach(d=>{const tr=document.createElement('tr');tr.innerHTML=`<td>${d.dep_id}</td><td>${d.dep_name}</td><td>${d.location||''}</td><td class="actions"><button class="btn ghost" onclick="deleteDepartment('${d.dep_id}')">Delete</button></td>`;tbody.appendChild(tr)}} catch(e){console.error(e)}}
```

```
async function deleteDepartment(id){if(!confirm('Delete department '+id+'?'))return;await fetch('/api/departments/'+id,{method:'DELETE'});fetchDepartments();flash('Deleted')}
```

```
async function searchDepartment(){const id=document.getElementById('search_dep_id').value.trim();if(!id){flash('No ID provided');return}const res=await fetch('/api/departments/'+id);const dep=await res.json();if(dep && dep.dep_id){alert('Found: ${dep.dep_name} — ${dep.location||'N/A'}')}else alert('Not found')}
```

```
async function updateDepartment(){const dep_id=document.getElementById('update_dep_id').value.trim();const dep_name=document.getElementById('update_dep_name').value.trim();if(!dep_id||!dep_name){flash('Provide ID and new name');return}await fetch('/api/departments/'+dep_id,{method:'PUT',headers:{'Content-Type':'application/json'},body:JSON.stringify({dep_name})});fetchDepartments();flash('Updated')}
```

// Courses

```
async function createCourse(e){e.preventDefault();const dep_id=document.getElementById('course_dep_id').value.trim();const c_id=document.getElementById('c_id').value.trim();const c_name=document.getElementById('c_name').value.trim();if(!dep_id||!c_id||!c_name){flash('Complete the fields');return}await fetch('/api/courses',{method:'POST',headers:{'Content-Type':'application/json'},body:JSON.stringify({dep_id,c_id,c_name})});fetchCourses();flash('Added')}
```

```
async function fetchCourses(){try {const res=await fetch('/api/courses');const data=await res.json();const tbody=document.querySelector('#courseTable tbody');tbody.innerHTML="";data.forEach(c=>{const tr=document.createElement('tr');tr.innerHTML=`<td>${c.c_id}</td><td>${c.dep_id}</td><td>${c.c_name}</td><td class="actions"><button class="btn ghost" onclick="deleteCourse('${c.c_id}')">Delete</button></td>`;tbody.appendChild(tr)}} catch(e){console.error(e)}}
```

```
async function deleteCourse(id){if(!confirm('Delete course '+id+'?'))return;await fetch('/api/courses/'+id,{method:'DELETE'});fetchCourses();flash('Deleted')}
```

```
async function searchCourse(){const id=document.getElementById('search_course_id').value.trim();if(!id){flash('No ID');return}const res=await fetch('/api/courses/'+id);const c=await res.json();if(c && c.c_id){alert('Found: ${c.c_name} — Dept: ${c.dep_id}')}else alert('Not found')}
```

```
async function updateCourse(){const c_id=document.getElementById('update_course_id').value.trim();const c_name=document.getElementById('update_course_name').value.trim();if(!c_id||!c_name){flash('Provide ID and new name');return}await fetch('/api/courses/'+c_id,{method:'PUT',headers:{'Content-Type':'application/json'},body:JSON.stringify({c_name})});fetchCourses();flash('Updated')}
```

// Modules

```
async function createModule(e){e.preventDefault();const c_id=document.getElementById('module_c_id').value.trim();const mod_id=document.getElementById('mod_id').value.trim();const mod_name=document.getElementById('mod_name').value.trim();if(!c_id||!mod_id||!mod_name){flash('Complete fields');return}await fetch('/api/modules',{method:'POST',headers:{'Content-Type':'application/json'},body:JSON.stringify({c_id,mod_id,mod_name})});fetchModules();flash('Added')}
```

```
async function fetchModules(){try {const res=await fetch('/api/modules');const data=await res.json();const tbody=document.querySelector('#moduleTable tbody');tbody.innerHTML="";data.forEach(m=>{const tr=document.createElement('tr');tr.innerHTML=`<td>${m.mod_id}</td><td>${m.c_id}</td><td>${m.mod_name}</td><td class="actions"><button class="btn ghost"
```

```
onclick="deleteModule('${m.mod_id}')">Delete</button></td>`;tbody.appendChild(tr))}}catch(e){console.error(e)}}

```

```
async function deleteModule(id){if(!confirm('Delete module '+id+'?'))return;await
fetch('/api/modules/'+id,{method:'DELETE'});fetchModules();flash('Deleted')}
```

```
async function searchModule(){const id=document.getElementById('search_mod_id').value.trim();if(!id){flash('No
ID');return}const res=await fetch('/api/modules/'+id);const m=await res.json();if(m && m.mod_id){alert('Found:
${m.mod_name} — Course: ${m.c_id}')}else alert('Not found')}
```

```
async function updateModule(){const mod_id=document.getElementById('update_mod_id').value.trim();const
mod_name=document.getElementById('update_mod_name').value.trim();if(!mod_id||!mod_name){flash('Provide ID and new
name');return}await fetch('/api/modules/'+mod_id,{method:'PUT',headers: {'Content-
Type':'application/json'},body:JSON.stringify( {mod_name} )});fetchModules();flash('Updated')}
```

```
// Wire up buttons
```

```
document.getElementById('export-dep-xlsx').addEventListener('click',()=>tableToCSV('departmentTable','departments'));
```

```
document.getElementById('export-course-xlsx').addEventListener('click',()=>tableToCSV('courseTable','courses'));
```

```
document.getElementById('export-mod-xlsx').addEventListener('click',()=>tableToCSV('moduleTable','modules'));
```

```
// Initial load
```

```
window.addEventListener('load',()=>{fetchDepartments();fetchCourses();fetchModules();});
```

```
</script>
```

```
</body>
```

```
</html>
```

Output/Result:

U

University Dashboard
overview — Departments · Courses · Modules

Connected to: /api/ (mock)

Departments

Create and manage academic departments

Refresh

Export XLSX

2

Department Name

Location

2

Search

2

BCA

Update

Add Department

ID	Name	Location	Actions
1	ML	Gurugram	Delete
2	BCA	Chandigarh university	Delete

Tip: Click Delete to remove a department. Use Update section to modify names.

Courses

Associate courses to departments

Refresh

Export XLSX

Department ID (eg. D001)

Course ID (eg. C101)

Course Name

Search Course ID

Search

ID to update

New name

Update

Add Course

ID	Dept ID	Name	Actions
----	---------	------	---------

Tip: Keep Dept ID consistent with Departments. Use Refresh after backend changes.

Modules

Course modules / subjects

Refresh

Export XLSX

Course ID (eg. C101)

Module ID (eg. M01)

Module Name

M123

Search

ID to update

New name

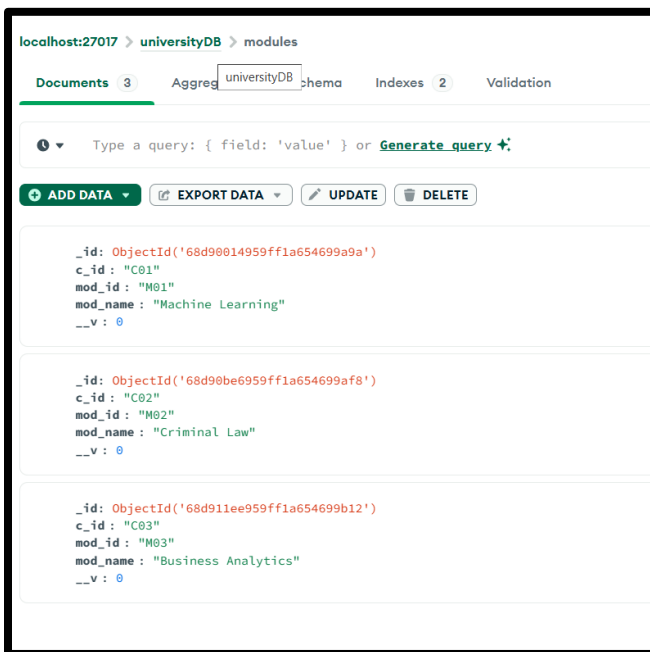
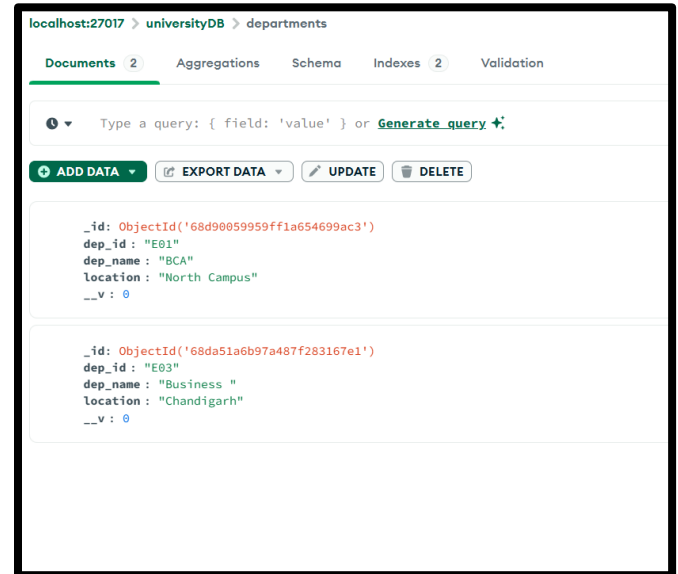
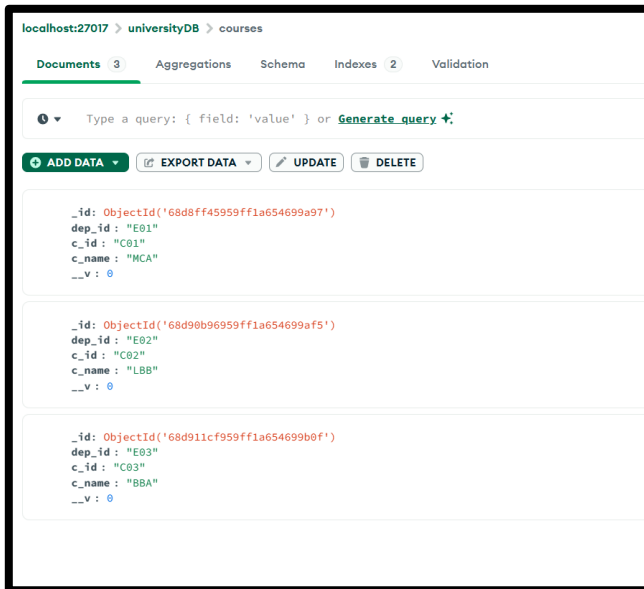
Update

Add Module

ID	Course ID	Name	Actions
M123	C123	Maths	Delete

Tip: Modules link to Course ID. Use Update to rename modules.

MongoDB:



Learning outcomes:

1. Ability to **design REST APIs** in Node.js/Express.
2. Skill to structure projects using **MVC pattern**.
3. Experience with **MongoDB schemas and relations**.
4. Confidence in **frontend-backend communication**.
5. Understanding of how to **document and present** a software project.